Geopolitical Risk & Financial Stress

Evidence from India

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Outline

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Motivation

- Geopolitical developments are increasingly important sources of macro-financial risk for emerging markets
- For India: border tensions, security incidents, and global strategic shifts affect growth expectations and financial conditions
- These shocks operate through risk premia and investor sentiment
- Standard macroeconomic indicators often fail to capture these effects

Research Gap

Existing Measures

- Financial Stress Index (FSI): Asian Development Bank's composite measure for India
- Caldara-Iacoviello GPR Index: Based on Western media and global events

Problem

Existing GPR measures don't adequately capture **India-specific** geopolitical dynamics

Our Solution

Construct India-specific geopolitical risk indices using local media sources and India-relevant keywords

Data Sources

- **Dependent Variable**: Monthly Financial Stress Index (FSI) from Park & Mercado (2014)
  - Period: 2010-2023 (156 observations)
  - Aggregates information from money markets, bonds, equities, FX, and banking
- **Independent Variables**: Custom GPR indices from LSEG data
  - $\sim$84,000 article headlines
  - Local Indian outlets: Hindustan Times, Indian Express, The Hindu, Economic Times, Mint, NDTV, BSE
  - India-specific keywords: India, Pakistan, Kashmir, LAC, LOC, China Border, Modi, terrorism, etc.

Index Construction - Definitions

Three Indices

- **GPT (Geopolitical Threat)**: war_threats, peace_threats, military_buildup, terrorist_threats
- **GPA (Geopolitical Act)**: war_beginning, war_escalation, terrorist_acts
- **GPR (Geopolitical Risk)**: any article tagged GPT or GPA

Methodology - 7 Step Process

1. **News Retrieval**: Collect headlines using India-related keywords
2. **Deduplication**: Remove duplicate stories based on unique IDs
3. **Exclusion Filtering**: Remove non-geopolitical content (cricket, bollywood, sports, movies)
4. **Classification**: Proximity search for topic-threat/act word pairs (within 0-2 words)
5. **GPT/GPA Designation**: Classify articles into threat or act categories
6. **Daily Share Calculation**: Daily index = (category articles) / (total articles)
7. **Normalization**: Scale to mean = 100 for comparability

Regression Model

Static OLS Specification

We estimate three separate regressions:

$$FSI_t = \beta_0 + \beta_1 GPR_t + \epsilon_t$$
$$FSI_t = \beta_0 + \beta_1 GPA_t + \epsilon_t$$
$$FSI_t = \beta_0 + \beta_1 GPT_t + \epsilon_t$$

where FSI_t is the Financial Stress Index and GPR_t, GPA_t, GPT_t are the geopolitical risk measures at time t.

Regression Results: All Indian Sources

Statistic	**GPR**	**GPA**	**GPT**
Coefficient (β_1)	0.0032	0.0019	0.0026
Std. Error	0.0016	0.0014	0.0012
t-statistic	1.9421	1.3145	2.0755
p-value	0.0539	0.1906	**0.0396***
R-squared	0.024	0.011	0.027
Adj. R-squared	0.018	0.005	0.021
F-statistic	3.772	1.728	4.308
Prob (F-stat)	0.0539	0.191	**0.0396***
Observations	156	156	156

Table 1: * indicates significance at 5% level

Regression Results: Local Sources Only

Statistic	**GPR**	**GPA**	**GPT**
Coefficient (β_1)	0.0028	0.0017	0.0019
Std. Error	0.0010	0.0009	0.0007
t-statistic	2.7763	1.9480	2.5710
p-value	**0.0062***	0.0533	**0.0111***
R-squared	0.049	0.025	0.043
Adj. R-squared	0.043	0.018	0.036
F-statistic	7.708	3.795	6.610
Prob (F-stat)	**0.0062***	0.0533	**0.0111***
Observations	151	151	151

Table 2: Core local outlets only; * significant at 5% level

Key Findings

1. GPT (Geopolitical Threats) - Significant

- Coefficient: 0.0026, p-value = 0.0396
- **Statistically significant at 5% level**
- Threats (war_threats, terrorist_threats) do affect financial stress

2. GPA (Geopolitical Acts) - Insignificant

- Coefficient: 0.0019, p-value = 0.1906
- Not statistically significant
- Actual events may already be priced into markets

3. GPR (Combined) - Marginally Significant

- Coefficient: 0.0032, p-value = 0.0539
- Marginally significant (close to 5% threshold)

Interpretation

- **Positive relationship**: Higher geopolitical threats $\rightarrow$ Higher financial stress
- **Threats matter more than acts**: Anticipation and uncertainty drive financial stress more than realized events
- **Low explanatory power**: R^2 at most 0.049 (at most 4.9% of FSI variation explained)
- **Implication**: Geopolitical risk is one factor among many affecting financial stress

Limitations (1/2)

1. Low Predictive Power
   - R^2 values very low (max 0.049, local sources)
   - Financial stress influenced by many factors beyond geopolitical risk
   - Missing: macroeconomic variables, global factors, monetary policy, market-specific dynamics

2. Reliance on Media-Based Measures
   - Subject to editorial biases and reporting cycles
   - Asymmetric attention to dramatic events
   - Sensitive developments may not be publicly covered

Limitations (2/2)

3. Subjectivity in Construction

- Manually curated keywords and sources
- No consensus on optimal keyword selection
- Different choices may yield different results

4. Static Linear Specification

- Contemporaneous OLS may miss dynamic/lagged effects
- More advanced models needed (VAR, local projections, GARCH)
- Policy responses and delayed capital flows not captured

5. FSI Measurement Issues

- FSI itself is a composite with measurement noise
- Reflects both domestic and global influences

Conclusion

Main Contributions

- Developed **India-specific** GPR, GPT, and GPA indices
- Used local Indian media sources and relevant keywords
- Filled gap left by Western-centric global GPR measures

Key Results

- **Geopolitical Threats (GPT)** significantly affect India's financial stress
- **Geopolitical Acts (GPA)** show no significant relationship
- Overall explanatory power remains low (R^2 at most 0.049)

Implication

While India-specific geopolitical threats matter for financial stress, they explain only a small fraction. A multifaceted approach incorporating additional economic and policy variables is needed.

Future Directions

1. Include additional control variables (GDP growth, inflation, oil prices, global risk factors)
2. Explore dynamic models (VAR, GARCH, local projections)
3. Test for lagged and lead-lag relationships
4. Analyze non-linear effects and regime-switching behavior
5. Extend to high-frequency (daily/weekly) analysis
6. Compare with other emerging markets

References

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Thank You!

Questions?